

Multi-label classification of Dutch election political manifestos

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Abstract

We study automatic thematic classification of Dutch election manifesto paragraphs in a multi-label setting, where each paragraph can receive multiple themes. The dataset comprises 2,533 paragraphs from the 1986, 1994, and 1998 elections, annotated with 380 themes and exhibiting a highly long-tailed label distribution. We compare classical one-vs-rest TF-IDF baselines across several text representations, dense sentence embeddings while keeping the best classical classifier fixed, and fine-tuning a Dutch transformer model (RobBERT-2023) using focal loss. On the full label space, the best TF-IDF pipeline (balanced Logistic Regression with lemmatised unigram+bigram features without stopwords) achieves a Micro-F1 of 0.508, while dense embeddings underperform due to very low precision despite high recall. Fine-tuned RobBERT-2023 improves over dense embeddings but remains below the classical baseline in the full long-tailed setting due to systematic over-prediction. In a controlled long-tail ablation, removing rare labels progressively improves performance for all models, with RobBERT-2023 surpassing the classical baseline once labels have sufficient training support. Overall, the results indicate that limited positive supervision per label appears to be a primary bottleneck for transformer-based multi-label theme classification in this dataset.

1 Introduction

Election manifestos provide key insights into how political parties present their priorities and positions to the public before an election. For Dutch elections, manifestos have been thematically indexed by experts to enable comparisons across parties and issues (Verberne et al., 2014). Such annotation supports political analysis and search tools. However, it does not scale well with new manifestos and emerging issues over time.

This study investigates automatic thematic classification of Dutch election manifesto paragraphs in a multi-label setting. Paragraphs often cover multiple themes, requiring classifiers to predict sets of labels rather than single categories. Severe label imbalance complicates the task. Frequent themes dominate while rare ones form a long tail, leading models to favour common labels at the expense of rare ones.

The research addresses four questions.

- 1. Classical Baselines:** How do classical non-transformer classifiers perform across different text representations in multi-label theme prediction?
- 2. Dense Embeddings:** Can dense sentence embeddings outperform sparse bag-of-words features when paired with strong classical baselines?
- 3. Transformers:** How do transformer-based models compare to the best classical approaches on this task?
- 4. Long-Tail Effect:** To what extent does the long tail of rare labels impact overall classification performance?

These questions quantify differences across model families, representations, and data characteristics to advance scalable manifesto analysis.

2 Background and Related Work

Text classification assigns predefined labels to text using supervised learning on labelled examples (Sebastiani, 2002). A strong and widely used baseline represents documents with sparse bag-of-words features, often weighted with TF-IDF, and trains linear classifiers. These methods remain competitive for topical tasks because they capture issue-specific vocabulary and scale well to high-dimensional feature spaces.

Our setting is multi-label classification, where a paragraph may receive several themes. A standard approach is binary relevance (one-vs-rest), which

Year	#Words	Vocabulary Size	#Paragraphs	#Themes	Words/para(std. dev.)	Themes/para(min-max, std. dev.)
Overall	802,796	59,567	2,533	380	317 (246)	8.5 (1–68, 6.0)
1986	289,735	29,759	783	214	370 (340)	10.6 (1–68, 8.1)
1994	268,284	29,764	935	217	287 (185)	6.9 (1–39, 4.2)
1998	244,777	28,617	815	218	300 (184)	8.4 (1–29, 4.7)

Table 1: Statistics of the manifesto paragraph dataset

trains one classifier per label and combines their outputs (Tsoumakas et al., 2007). Multi-label political data is typically highly imbalanced: a small number of themes occur frequently, while many appear only rarely. This long-tail structure makes it difficult to learn reliable decision boundaries for infrequent labels and can reduce performance on rare themes.

For Dutch election manifestos, Verberne et al. (2014) study automatic thematic classification with a large, fine-grained theme set and treat the task explicitly as multi-label classification. They report challenges that are characteristic of this domain, including skewed label frequencies, limited training data for many themes, and temporal change where topics evolve and new themes emerge over time.

More recent work on manifestos explores models beyond bag-of-words. For example, Zirn et al. (2016) investigates topic classification and topic shifts in manifestos, highlighting that themes can vary across segments and across time. Neural approaches have also been applied to manifesto analysis, including architectures that combine sentence-level and document-level signals (Subramanian et al., 2017). These developments motivate comparing sparse lexical representations with denser semantic representations.

Transformer models are now widely used for text classification. For Dutch, widely used monolingual transformer models include BERTje (2019) (de Vries et al., 2019), a RoBERTa-based Dutch model 2020 (Delobelle et al., 2020), and the updated RobBERT-2023 release, which is trained on OSCAR 2023 with an updated tokeniser via model conversion (Delobelle and Remy, 2024). In imbalanced multi-label settings, losses designed to reduce the dominance of easy negatives are often considered; focal loss is a well-known example (Lin et al., 2017). In this assignment, we perform controlled comparisons between classical TF-IDF baselines, dense sentence embeddings, and a Dutch transformer model, and we analyse how removing rare labels changes performance in the long tail.

3 Data

We use a dataset of Dutch election manifestos that has been segmented into paragraphs and annotated with thematic labels. Each instance in our experiments is one paragraph (a `<p>` segment) and the prediction target is a set of one or more themes. This makes the task a multi-label classification problem rather than a single-label topic assignment.

Annotation structure presents two key challenges. First, paragraphs are label-dense and frequently address multiple policy issues at once. Second, the label set is large and imbalanced: a small number of themes occur frequently, while many themes appear only a few times. This long-tail distribution makes it difficult to learn reliable decision boundaries for rare labels and motivates the long-tail analysis in our fourth research question.

3.1 Dataset statistics

Table 1 summarises the dataset at the paragraph level, both overall and per election year. In total, we extracted 2,533 paragraphs with 380 unique themes. Paragraphs have an average length of 317 words (std. 246). The annotations are dense: paragraphs contain 8.5 themes on average and can contain as many as 68 themes.

Figure 1 shows the frequency of the 50 most common themes. The distribution is highly skewed: a small number of themes dominates the corpus (e.g., *milieubeleid* and *werkgelegenheid*), while most themes occur far less often. This long-tail imbalance affects evaluation and model choice, since a system can score well on frequent themes while still failing on rare labels.

4 Methods

This section describes the modelling approach for multi-label thematic classification of Dutch election manifesto paragraphs, including the task formulation, text representations, model families, and prediction procedures. The experimen-

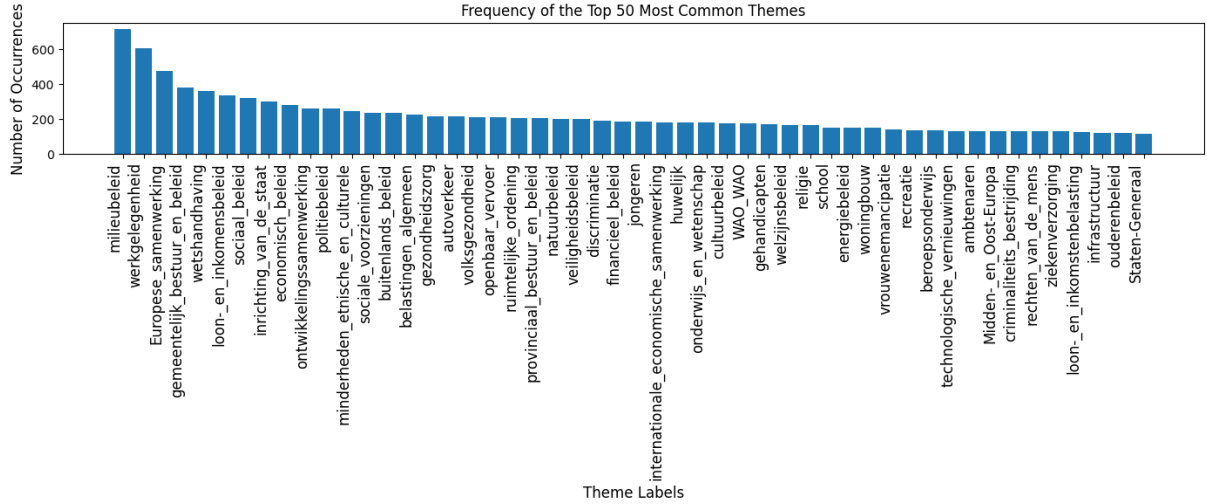


Figure 1: Frequency of the 50 most common themes in the Dutch Political Election Manifesto dataset

tal setup—including data splitting, hyperparameter selection, label filtering, and evaluation metrics—is presented in Section 5.

4.1 Task formulation

We formulate thematic coding of Dutch election manifesto text as a multi-label text classification problem. Each instance is a single paragraph, and the output is a set of one or more theme labels. For classical approaches, we adopt a one-vs-rest (binary relevance) formulation, in which a separate binary classifier is trained for each label, and predictions are combined into the final label set. Since the label distribution is highly imbalanced (Section 3), we additionally study how excluding very rare labels affects performance.

4.2 Text representations and classical models

We compare several sparse bag-of-words representations using TF-IDF weighting. All variants start from the same lowercased paragraph text but differ in token normalisation and n-gram inclusion:

- Unigrams.
- Lemmatised unigrams.
- Lemmatised unigrams with stopword removal.
- Unigrams + bigrams.
- Lemmatised unigrams + bigrams.
- Lemmatised unigrams + bigrams with stopword removal.

Each representation is evaluated with the same set of classical classifiers: Multinomial Naïve Bayes, Logistic Regression, Logistic Regression with balanced class weights, and linear Support Vector Machine (LinearSVC). We compare 24 classical pipelines (4 classifiers $\times$ 6 TF-IDF representations) and select the best-performing pipeline based on test-set Micro-F1, using Macro-F1 as a tie-breaker. This selected pipeline is used as the reference baseline in subsequent experiments.

4.3 Dense embedding comparison

To test whether dense semantic representations improve over sparse TF-IDF features, we encode each paragraph into a fixed-length dense vector using a pretrained SentenceTransformer model applied to the raw paragraph text. We then train the same best-performing classical classifier (with the same hyperparameters) on these dense vectors in a one-vs-rest setup. This comparison isolates the effect of feature representation while holding the classifier family constant.

4.4 Transformer-based classification

We fine-tune RobBERT-2023 as a Dutch-specific transformer baseline for manifesto text. Compared to earlier RobBERT, it uses a newer tokeniser and a more recent Dutch pretraining corpus. Paragraphs are tokenised with the model’s subword tokeniser and truncated/padded to 512 tokens.

The model outputs one logit per label, which is converted to probabilities via the sigmoid function. Labels are predicted independently (no soft-

max over mutually exclusive classes). To mitigate the dominance of easy negatives in an imbalanced label space, we optimise the model using focal loss. At inference, we tune a single global threshold on the validation set by sweeping candidate values from 0.01 to 0.91 in steps of 0.025 and selecting the threshold that maximises macro-F1. This prioritises balanced performance across labels, which is essential given the long tail of rare themes. Final performance is reported on the test set using the tuned threshold.

4.5 Long-tail analysis

We analyse the impact of the long-tail label distribution by progressively removing rare labels using minimum global frequency cutoffs (5, 25, 50, 75, and 100 occurrences). For each cutoff, we re-train and evaluate both (i) the best classical pipeline and (ii) the RobBERT-2023 model on the filtered label set. Labels that become untrainable in the training split after filtering are removed to ensure stable one-vs-rest training. This controlled ablation shows how performance changes as label imbalance is systematically reduced.

5 Experimental Setup

This section specifies the experimental protocol used throughout the study to ensure reproducibility and comparable results. We describe the data splitting strategy, label filtering procedure, evaluation metrics, and key hyperparameters. The setup remains consistent across all models, representations, and analyses.

5.1 Data Splitting

We split the dataset into training, validation, and test sets using a multi-label stratified procedure to preserve label co-occurrence patterns across splits. The split proportions are fixed at 70% training, 15% validation, and 15% test, and remain constant across all experiments.

5.2 Label filtering

To prevent training instability in one-vs-rest classifiers, we remove label columns with fewer than two positive examples in the training set. The same filtered label set is applied consistently to the training, validation, and test targets. In the long-tail analysis, this filtering is re-applied after each cutoff, and instances that would otherwise have zero remaining labels are discarded.

5.3 Evaluation measures

We report micro-averaged precision, recall, and F1-score, together with macro-averaged F1-score. Micro-averages are computed by aggregating true positives (TP), false positives (FP), and false negatives (FN) over all labels and all instances in the test set, which gives more influence to frequent themes. Macro-F1 is computed by first calculating an F1-score independently for each label and then taking the unweighted mean across labels, making it more sensitive to performance on rare themes.

The micro-averaged measures are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{Micro-F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

Macro-F1 is defined as:

$$\text{Macro-F1} = \frac{1}{L} \sum_{i=1}^L F1_i, \quad (4)$$

where L is the number of labels and $F1_i$ is the per-label F1-score for label i . For labels where precision or recall is undefined due to zero predicted or zero true positives, we follow the implementation used in our experiments and set the corresponding value to zero.

5.4 Hyperparameters

We fix the random seed to 42 for consistency across experiments. TF-IDF representations use `max_features=5000`; the n-gram range is set to (1,1) for unigram variants and (1,2) for unigram+bigram variants. For RobBERT-2023, we use a maximum sequence length of 512 tokens, batch size 2 with gradient accumulation steps of 4, learning rate $2e-5$, 10 epochs, a cosine learning rate schedule with warmup ratio 0.1, and focal loss with $\gamma = 2.0$ and $\alpha = 0.25$. The long-tail analysis applies global frequency cutoffs of 5, 25, 50, 75, and 100 occurrences.

6 Results

This section presents test-set performance for the classical baselines, the comparison of dense embeddings and the transformer model, and the long-

Table 2: Classical one-vs-rest baselines across TF-IDF representations

Representation	Model	Precision	Recall	Micro-F1	Macro-F1
Unigrams	Naive Bayes	0.889	0.016	0.032	0.003
	Logistic Regression	0.908	0.058	0.109	0.015
	Logistic Regression (balanced)	0.447	0.580	0.504	0.329
	Linear SVM	0.805	0.221	0.347	0.128
Unigrams lemmatised	Naive Bayes	0.854	0.012	0.024	0.002
	Logistic Regression	0.896	0.058	0.109	0.016
	Logistic Regression (balanced)	0.425	0.590	0.494	0.326
	Linear SVM	0.796	0.222	0.347	0.126
Unigrams lemmatised (no stopwords)	Naive Bayes	0.898	0.034	0.065	0.006
	Logistic Regression	0.889	0.082	0.151	0.026
	Logistic Regression (balanced)	0.457	0.570	0.508	0.347
	Linear SVM	0.782	0.250	0.379	0.155
Unigrams + Bigrams	Naive Bayes	0.895	0.030	0.058	0.005
	Logistic Regression	0.893	0.056	0.105	0.015
	Logistic Regression (balanced)	0.444	0.558	0.495	0.310
	Linear SVM	0.793	0.212	0.334	0.113
Unigrams + Bigrams lemmatised	Naive Bayes	0.864	0.026	0.051	0.004
	Logistic Regression	0.901	0.056	0.106	0.015
	Logistic Regression (balanced)	0.430	0.556	0.485	0.303
	Linear SVM	0.788	0.209	0.330	0.111
Unigrams + Bigrams lemmatised (no stopwords)	Naive Bayes	0.876	0.043	0.083	0.009
	Logistic Regression	0.898	0.083	0.152	0.028
	Logistic Regression (balanced)	0.459	0.569	0.508	0.344
	Linear SVM	0.790	0.254	0.385	0.161

tail ablation. Results are reported using micro-averaged precision, recall, and F1-score, together with macro-averaged F1-score.

6.1 Overall Performance of Classical Baselines

Table 2 summarises the performance of classical one-vs-rest classifiers across six TF-IDF representations (unigrams/bigrams, lemmatised or not, with/without stopwords). Among the evaluated classical pipelines, the strongest configuration is Logistic Regression with balanced class weights on lemmatised unigrams and bigrams without stopwords, achieving Precision=0.459, Recall=0.569, Micro-F1=0.508, and Macro-F1=0.344 on the test set. This pipeline serves as the reference baseline for subsequent comparisons.

6.2 Comparison of Dense Embeddings and Transformer Model

Table 3 compares the selected TF-IDF reference pipeline to two alternatives. For dense embeddings, we keep the classifier family fixed (balanced Logistic Regression in one-vs-rest form)

and replace TF-IDF features with paragraph-level sentence embeddings. This embedding setup yields higher recall (0.603) but much lower precision (0.178), resulting in lower Micro-F1 (0.275) and Macro-F1 (0.190) than TF-IDF. RobBERT-2023 performs better than dense embeddings (Micro-F1 = 0.310) but remains below the TF-IDF reference system overall, with Macro-F1 = 0.162.

Table 3: Summary of TF-IDF (best), dense embeddings and RobBERT-2023 results

Method	Precision	Recall	Micro-F1	Macro-F1
TF-IDF (best)	0.459	0.569	0.508	0.344
Dense embeddings	0.178	0.603	0.275	0.190
RobBERT-2023	0.202	0.666	0.310	0.162

6.3 Long-tail analysis

Table 4 shows performance when labels with global frequency below a cutoff are removed. For both the best classical pipeline and RobBERT-2023, Micro-F1 and Macro-F1 increase monotonically with higher cutoffs. At cutoff 100 (57 remaining labels), the classical system achieves Micro-F1=0.593 and Macro-F1=0.562,

Table 4: Long-tail analysis: performance as labels with global frequency below the cutoff are removed.

Cutoff	Labels	Train	Test	Classical		RobBERT-2023	
				Micro-F1	Macro-F1	Micro-F1	Macro-F1
5	354	1759	399	0.509	0.360	0.370	0.177
25	202	1751	397	0.530	0.472	0.448	0.360
50	130	1725	392	0.546	0.497	0.513	0.447
75	81	1699	389	0.571	0.530	0.581	0.533
100	57	1664	384	0.593	0.562	0.633	0.598

while RobBERT-2023 reaches Micro-F1=0.633 and Macro-F1=0.598. This indicates that reducing the long-tail improves both overall and balanced performance, with RobBERT-2023 gaining relatively more on macro-F1 and surpassing the classical baseline from cutoff 75 onwards.

7 Discussion

This section interprets the experimental findings in relation to multi-label thematic classification of Dutch manifesto paragraphs. We focus on what the results reveal about feature representations, model families, and the role of label imbalance.

7.1 Classical baselines and representation choices

The classical TF-IDF experiments show that the best-performing configuration is Logistic Regression with balanced class weights on lemmatised unigrams and bigrams without stopwords, achieving Micro-F1 = 0.508 and Macro-F1 = 0.344. Models without imbalance correction (standard Logistic Regression and Naïve Bayes) achieve very high precision but extremely low recall, indicating that they predict few labels per paragraph and therefore miss many gold themes.

Applying `class_weight=balanced` substantially improves this trade-off by increasing recall while maintaining usable precision, resulting in the strongest overall performance among classical models. Linear SVM also attains high precision but lower recall, suggesting a stricter decision boundary under the default one-vs-rest setup.

Across representations, lemmatisation and bigrams yield modest gains, likely by reducing sparsity and capturing informative short phrase patterns. In the best-performing setting, stopword removal further improves performance, consistent with the limited thematic value of frequent function words in TF-IDF representations.

7.2 Dense embeddings versus sparse TF-IDF

Dense sentence embeddings perform worse than the TF-IDF reference system. They achieve higher recall (0.603) but very low precision (0.178), which reduces Micro-F1 to 0.275 and Macro-F1 to 0.190. In other words, the embedding setup predicts many extra themes per paragraph, and most of those additional labels are false positives. For this dataset, sparse TF-IDF features appear to be better aligned with the task because many themes are signalled by specific words and short phrases that TF-IDF represents directly.

7.3 Transformer fine-tuning

Fine-tuned RobBERT-2023 improves over the dense embedding baseline (Micro-F1 0.310 vs. 0.275) but remains below the TF-IDF reference system (0.310 vs. 0.508). The key difference is precision: RobBERT-2023 has much lower precision (0.202) while recall is higher than TF-IDF (0.666 vs. 0.569). This indicates over-prediction: the model assigns many themes that are not present in the gold labels, so the recall gains do not fully translate into a higher F1 score.

Macro-F1 is particularly low for RobBERT-2023 (0.162), which suggests weak performance on infrequent labels. Given the large label space (380 themes) and the long-tailed frequency distribution, many labels have very few positive training examples, which limits how well a transformer can learn label-specific decision boundaries. We also use one global probability threshold for all labels (tuned on validation to maximise Macro-F1). A single threshold is simple and consistent, but it cannot adapt to the fact that frequent and rare labels typically require different operating points; label-specific thresholds could improve the precision-recall balance.

7.4 Long-tail effects and rare label removal

The long-tail analysis highlights the impact of label imbalance. As labels below a global frequency

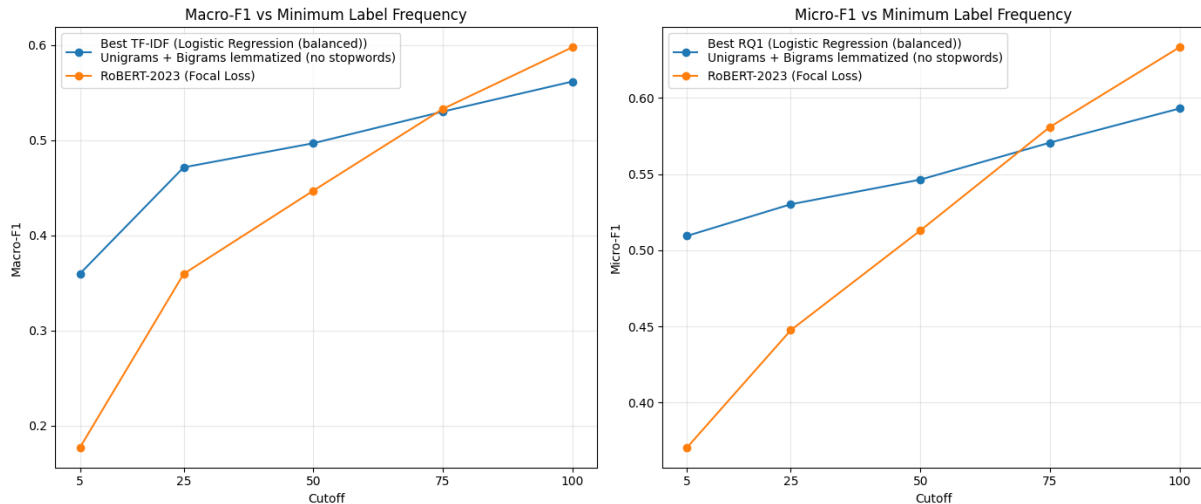


Figure 2: Long-tail analysis: Micro-F1 and Macro-F1 under increasing minimum label-frequency cutoffs

cutoff are removed, performance improves for both systems, with RobBERT-2023 gaining relatively more. At cutoff 5, the classical baseline still leads (Micro-F1 0.509 vs. 0.370), but from cutoff 75 onward RobBERT-2023 overtakes it. As shown in Figure 2, at cutoff 100 (57 labels) RobBERT-2023 achieves Micro-F1 = 0.633 and Macro-F1 = 0.598.

This trend suggests that the transformer’s main limitation in the full setting is not architectural capacity but the scarcity of training examples per label. When restricted to more frequent themes, RobBERT-2023 can better exploit contextual representations and generalise effectively. The classical baseline also improves, but more gradually, indicating that sparse lexical cues remain robust even with moderate imbalance.

7.5 Overall comparison across model families

On the full label set, the best classical TF-IDF pipeline with balanced Logistic Regression achieves the strongest overall performance, outperforming both dense embeddings and fine-tuned ROBBERT-2023. Dense embeddings and the transformer model both suffer from low precision in the long-tailed setting, indicating systematic over-prediction, although ROBBERT-2023 improves substantially as rare labels are removed.

Overall, the results show that limited positive supervision per label is the primary bottleneck for fine-grained multi-label theme prediction, even for large pretrained transformers, as reflected by consistent performance gains when very low-frequency labels are excluded.

7.6 Implications for manifesto analysis

For practical manifesto coding across a large and highly imbalanced theme inventory, a classical TF-IDF approach provides a robust and computationally lightweight solution. It delivers the most reliable performance when full thematic coverage is required, including rare themes.

Transformer fine-tuning becomes more attractive when analysis is restricted to well-represented themes, or when combined with extensions that explicitly address rare-label challenges, such as label-specific decision thresholds or methods that increase effective supervision for infrequent themes. These findings suggest that model selection for manifesto analysis should be guided less by architectural complexity and more by label frequency characteristics and the intended scope of thematic coverage.

7.7 Comparison to prior work

Our findings align closely with the observations of Verberne et al. (2014), who studied the same Dutch manifesto corpus and explicitly formulated thematic coding as a multi-label classification problem. Verberne et al. identify strong label imbalance and limited training data per theme as central challenges, and report that lexical, linear classifiers achieve relatively high precision but struggle to achieve high recall across the full theme set. We observe a comparable precision-recall trade-off for classical TF-IDF models, and show that applying imbalance-aware weighting substantially improves recall while maintaining competitive precision.

A key extension of our work is a controlled long-tail analysis that quantifies the impact of label frequency on model performance. Whereas Verberne et al. discuss the effects of scarce supervision and evolving themes qualitatively, our experiments demonstrate this effect directly by progressively removing rare labels. Across all model families, performance improves as very infrequent themes are excluded. In particular, RobBERT-2023 overtakes the classical TF-IDF baseline once labels have sufficient training support, as shown by higher Micro-F1 and Macro-F1 scores at higher frequency cutoffs. These results provide empirical support for the conclusion that limited positive examples per theme are a major constraint in fine-grained manifesto classification, and they extend prior work by making the role of label frequency explicit and measurable.

8 Conclusion

We studied multi-label thematic classification of Dutch election manifesto paragraphs using classical TF-IDF pipelines, dense sentence embeddings, and a fine-tuned Dutch transformer model (RobBERT-2023). On the full label space, the best overall performance was obtained by a TF-IDF system with balanced Logistic Regression. Dense embeddings under the same classifier family underperformed, mainly due to very low precision despite high recall. Fine-tuned RobBERT-2023 improved over dense embeddings but did not surpass the best classical system in the full long-tailed setting, suggesting systematic over-prediction and limited supervision for many rare themes.

The long-tail analysis shows that performance improves for both model families as rare labels are removed, and that RobBERT-2023 benefits most when labels have sufficient training support: it overtakes the classical baseline from cutoff 75 onward and achieves the best results at higher cutoffs. Future work could test label-specific thresholds and methods that explicitly model label dependencies, which may further improve performance on infrequent themes.

9 Reflection

Luis handled the classical and embedding models; Nallathambi handled RobBERT-2023 and long-tail experiments. Both of us jointly finalised the report.

We used ChatGPT AI for three limited tasks:

exploring recent related work as an alternative to manual literature searches, surveying Dutch transformer models, and fixing minor LaTeX table-formatting issues. The assistant was helpful but not fully reliable: suggested literature was manually verified and shortlisted, while formatting fixes were accurate and time-saving.

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